

Claire Vlases

Professor Schroeder

Science, Values, and Democracy: Final Paper

13 Dec 2024

Navigating Politicization: Reconciling Douglas and Lusk's Frameworks

The politicization of science, especially in a high-stakes context such as climate change and public health, raises critical questions about the boundaries between legitimate political engagement and the illegitimate politicization of scientific inquiry. As Heather Douglas observes, “Increasingly, there is concern that the ability of science to ‘speak truth to power’ has been eroded by political interference in science or political forces tampering with science” (Douglas 123). At the same time, democratic governance requires scientific knowledge to make informed policy decisions. These decisions involve moral, social, and political non-epistemic values. The challenge lies not in determining whether such values influence science (they inevitably do), but in understanding when these values undermine the epistemic integrity upon which science depends.

The paper examines the boundaries between legitimate and illegitimate political influence on scientific inquiry must be examined in order to adjudicate between Douglas and Greg Lusk's frameworks that address this tension. Douglas argues that politicization occurs when political levers improperly infiltrate the norms of scientific inquiry, undermining its epistemic integrity. Alternatively, Lusk emphasizes the importance of democratic legitimacy in the selection of non-epistemic values that influence science, suggesting that politicization arises when these values are undemocratically imposed. While each framework offers valuable insights into the nature of politicization, neither fully resolves the challenge of balancing the epistemic demands of

science with sciences' independent societal responsibilities. Arguably, politicization occurs under the conditions outlined by Douglas and Lusk: first, when political norms infiltrate the distinctive space of scientific inquiry, violating its epistemic integrity; and second, when value selection lacks democratic legitimacy. To ensure a complete understanding, the best practice is a hybrid framework that synthesizes Douglas's and Lusk's insights, preserving the distinct norms of inquiry and that the values influencing science are democratically selected.

I. Rejection of the VFI

For much of its history, science has been regarded as objective, with the primary goal of uncovering knowledge and truth about the world. However, as Arnon Keren notes in reference to philosopher Philip Kitcher, "to the extent that science is aimed at the attainment of truth, it is not aimed at the attainment of all truths, but only of significant truths" (Kitcher 2001 as cited by Keren 235). These *significant truths* are often shaped by personal and societal priorities, allowing non-epistemic values to influence scientific inquiry. As Lusk observes, "a near consensus has emerged among philosophers of science that non-epistemic values have a permissible role to play in many core aspects of science" (Lusk 102). While the inclusion of non-epistemic values in science can be beneficial—Douglas even asserts that "social and ethical values are central to the responsible practice of scientific inquiry" (Douglas 125)—their integration must be approached with caution. The challenge lies in distinguishing legitimate political influences that enrich inquiry from those that compromise its foundational norms and lead to political influence.

Despite Lusk's assertion of a "near consensus" on the role of non-epistemic values in science, some philosophers of science maintain that scientific inquiry can and should remain insulated from non-epistemic values. This claim, known as Value-Free Ideal (VFI), is neither

realistic nor desirable, especially in a context when the consequences of scientific error have profound societal implications. As Douglas explains, “For science that has clear non-epistemic impacts, being ‘value-free’ is not a laudable goal” (Douglas 560). While the rejection of the VFI is widely accepted, it does not imply that all non-epistemic values should influence science. Moreover, the mere presence of non-epistemic values in scientific inquiry does not inherently constitute politicization. The rejection of the VFI simply opens the door to understanding that non-epistemic values play a role in science, and their inclusion carries a potential risk of politicization.

II. Douglas’s Framework: Understanding and Critique

Douglas offers a framework for understanding politicization that focuses on the distinctiveness of scientific inquiry. She defines the politicization of science “as an illegitimate political influence on science” (Douglas 123). For Douglas, maintaining epistemic integrity requires that the norms governing scientific inquiry remain distinct from those of democratic politics. To illustrate this distinction, Douglas examines norms that appear to be shared between science and politics but explains how they function differently in each domain, thereby separating the spaces for science and politics. She identifies four key norms that define the space of inquiry: clear and public restraints, criticism and response, fostering diversity, and open-ended debate (Douglas 128).

For example, the norm of criticism and response requires all participants to engage in reasoned critique and rebuttal. While this norm may appear superficially similar to practices in democratic politics, where criticism is also present, Douglas highlights a key difference. In politics, criticism does not necessitate a response, and often serves as a form of theater rather than

a genuine attempt to convince someone to change their viewpoint (Douglas 135). In contrast, within the space of inquiry, participation in criticism and response is mandatory. As Douglas notes: “Failing to participate in criticism and response is a reason to be excluded from the space of inquiry” (Douglas 135). In politics, abstention from criticism may be tolerated and ideological loyalty promoted, but these practices undermine the norms of criticism and diversity that are vital for a robust inquiry.

However, while Douglas’s framework emphasizes the distinctiveness of inquiry, it does not fully address how non-epistemic values, which inevitably influence science, should be selected or justified. This limitation leaves unanswered questions about how scientific priorities can remain democratically accountable while adhering to the norms of inquiry.

Moreover, Douglas’s view risks isolating scientific inquiry from the democratic processes that shape societal priorities. “Damaging politicization,” she writes, “occurs when democratic political norms are taken to be appropriate within the space of inquiry, overriding how some norms within the space of inquiry should function” (Douglas 124). Douglas’s view is necessary for separating politics and science, but it raises concerns about how or when democratically values can legitimately influence scientific priorities without undermining leaking into the space for inquiry. Scientific research does not occur in a vacuum; decisions about funding, resource allocation, and research focus are inherently tied to societal needs and priorities. If these non-epistemic values are not selected through inclusive and transparent democratic processes, science risks becoming disconnected from the communities it aims to serve, fostering public mistrust and reducing its societal impact.

III. Lusk’s Model

Lusk approaches the problem of politicization from a different angle, focusing on the processes that legitimize the inclusion of non-epistemic values in science. For Lusk, the presence of non-epistemic values is similarly not inherently problematic; rather, politicization arises when these values are selected in ways that fail to meet the standards of democratic legitimacy (Lusk 110). Central to Lusk's framework is the concept of deliberative democracy, which emphasizes the role of equality, reciprocity, absence of coercion, and fairness in decision-making processes (Lusk 107). Deliberative democracy demands that participants justify their positions with arguments that others can reasonably accept. As Lusk explains, "Democratic decisions based on value-laden science are politically legitimate to the degree that the value selection process upheld deliberative democratic ideals" (Lusk 108). This approach aligns the inclusion of values in science with democratic norms, ensuring that scientific priorities reflect societal needs without being influenced by power imbalances or ideological biases of a single entity. Lusk's framework is particularly valuable in addressing the societal dimensions of politicization. By grounding value selection in deliberative democratic processes, this framework ensures that scientific practices are accountable to diverse stakeholders and not merely the biases of elites, elected officials, or special interests, particularly when the scientific research is funded by an entity seeking a particular scientific outcome.

However, Lusk's focus on democratic legitimacy faces challenges in preserving the epistemic rigor of scientific inquiry. While his framework ensures inclusivity and accountability, it does not fully address the distinct norms of inquiry that Douglas highlights as essential for reliable scientific data. For example, the requirement to justify values democratically may conflict with the epistemic norms of criticism and response, particularly when political participants lack expertise or willingness to engage in rigorous critique.

IV. Combining Forces

While Douglas and Lusk offer compelling frameworks for addressing politicization, neither alone squarely resolves the tension between safeguarding scientific inquiry and ensuring its accountability and vetting essential to the democratic process. Reconciling these perspectives requires a hybrid approach that acknowledges the strengths of both frameworks. Lusk's deliberative democracy can guide the selection of non-epistemic values that influence science, ensuring these values are democratically legitimate and reflective of societal needs. However, once values are selected, Douglas's inquiry norms must govern their implementation to preserve the epistemic rigor of scientific practice. This approach creates a clear division of labor: deliberative democracy legitimizes value selection, while the norms of inquiry safeguard how those values are operationalized within science.

Combining Lusk's deliberative democracy with Douglas's norms of inquiry offers clear benefits, but also potential drawbacks. One concern is the tension between democratic inclusivity and the epistemic rigor required in inquiry. Deliberative democracy may legitimize values that conflict with scientific norms—such as ideological commitments that resist critique or evidence-based revision. Additionally, requiring scientists to respond to work shaped by undemocratic values could burden the space of inquiry with addressing pseudoscience or politically motivated research, diverting resources from legitimate critique. Balancing the demands of democratic accountability with the need to safeguard the epistemic integrity of science is a complex challenge that could create friction between the two frameworks.

However, the hybrid approach addresses the risks inherent in both frameworks. Lusk's principles prevent the imposition of undemocratic or ideologically biased values, while Douglas's

norms protect inquiry from being co-opted by political loyalty or dogma. Together, they offer a comprehensive strategy for navigating the complex interplay between science and politics, ensuring that scientific inquiry remains both epistemically rigorous and democratically accountable.

V. Case Studies

The challenges of politics are particularly noticeable in high-stakes domains where scientific inquiry and political decision-making intersect. The following cases demonstrate the necessity of integrating Douglas's and Lusk's frameworks to address both the need to maintain epistemic rigor and the inevitability of non-epistemic values shaping research priorities and outcomes.

Case Study 1: Looks Politicized, Is Politicized

The 1996 Dickey Amendment exemplifies illegitimate politicization in science. Prompted by accusations from the National Rifle Association (NRA) that the Centers for Disease Control and Prevention (CDC) was biased against guns, the NRA began lobbying Congress to take action (Rostron 2018). The amendment explicitly prohibited the CDC from using funds to "advocate or promote gun control," but its broader, tacit impact effectively stifled public health research on gun violence (Omnibus, as cited by Rostron 2018). Fearing political backlash or the loss of funding, researchers avoided the topic altogether, leading to a chilling effect on an entire field of inquiry.

The Dickey Amendment fails Douglas's norm of clear and public restraints, creating a chilling effect on an entire field of inquiry. A better implementation under Douglas's framework would involve explicit, narrowly targeted constraints on research funding that are publicly

debated and justified. For instance, Congress could clarify that funds may not be used to advocate for specific gun policies while still supporting broad public health research on gun violence. Such clarity would preserve the open-ended nature of scientific inquiry while respecting democratically determined boundaries.

Moreover, in order to transition such a politicized area towards less politicization, Lusk's deliberative democracy could provide a framework to address the selection of non-epistemic values in such cases. For example, the societal need to protect public safety could justify some prioritization of gun violence research. In a deliberative democratic process, stakeholders—including researchers, public health officials, and community representatives— could openly converse about how values like public safety and individual rights should shape research priorities. This inclusivity ensures that the selected values reflect broad societal needs rather than narrow political agendas.

This case also shows the importance of using Lusk and Douglas together. Although this example is far from the kind of deliberative democracy envisioned by Lusk, the Dickey Amendment was passed through a democratic system. Its creation shows the limitations of relying solely on democratic mechanisms to prevent polarization. Even if the values of public safety and individual rights had been deliberately selected in a Luskian framework, his model cannot safeguard against the epistemic risks posed by vague, coercive, or overly broad implementation of those values. Without Douglas's norms of inquiry, even values chosen through a democratic process risk being operationalized in ways that compromise the integrity and efficacy of scientific research.

Luckily, the Dickey Amendment was revised with clearer language after twenty years. Recent data shows that from 2020 to 2022, following the restart of congressional appropriations,

the CDC and NIH awarded approximately \$49 million and \$100.5 million, respectively, for firearm injury prevention research. During this same period, the number of registered clinical trials and research publications increased dramatically—by 90% and 86%, respectively, compared to 2017–2019 (Lin 2022). These outcomes highlight how clearer guidelines and funding support, informed by both democratic legitimacy and epistemic rigor, can rejuvenate stalled research fields and foster meaningful societal impact.

Case Study 2: Does Not Look Politicized, Is Not Politicized

The Human Genome Project (HGP) provides a compelling example of a case that does not appear politicized and, upon analysis, is not. Launched in 1990 as an international collaboration to map the entirety of human DNA, the HGP's goals were ostensibly scientific and epistemic, prioritizing the advancement of biological understanding and the improvement of public health (Genome 2024). Importantly, the project maintained its epistemic integrity by adhering to clear norms of inquiry while incorporating non-epistemic values that were democratically legitimate and reflective of societal needs. It demonstrates how the model of Lusk's deliberative democracy, paired with Douglas's norms of inquiry, can guide value selection and implementation in scientific research.

In the case of the HGP, the values driving the research—improving human health, understanding genetic diseases, and advancing scientific knowledge—were broadly endorsed through democratic mechanisms (Gannett 2023). Deliberative processes were evident in the ethical guidelines and policy discussions surrounding the project, such as how to handle sensitive data and avoid genetic discrimination. These discussions ensured that the non-epistemic values informing the HGP reflected societal priorities rather than narrow political or economic interests.

Once these values were democratically selected, Douglas's inquiry norms were also in place. The project made its methodologies and progress publicly available, upholding clear and public restraints by explicitly defining its scope, goals, and ethical boundaries. It also adhered to the norm of criticism and response by inviting regular peer review and engaging with critiques from the scientific community and ethicists. Furthermore, the HGP embraced the norm of fostering diversity by incorporating international collaboration and addressing global concerns, such as the equitable sharing of genetic information and the implications for developing nations.

Case Study 3: Seems Politicized, Is Not Politicized

The Montreal Protocol, adopted in 1987 to phase out substances depleting the ozone layer, was initially perceived as highly politicized. Critics argued that the science that informed the protocol was influenced by environmental lobbying and industrial resistance, with enforcement mechanisms reflecting political agendas rather than purely scientific considerations (Grey 2018). While the protocol is inherently political, that does not make the science behind it *politicized*. Although it involved significant economic and regulatory implications, its foundation in rigorous science and democratically selected values demonstrates otherwise (DeSombre 2000). Upon closer inspection, the protocol's success was grounded in epistemically rigorous research and inclusively selected societal values. It exemplifies how Lusk's deliberative democracy and Douglas's inquiry norms can work together to address concerns about perceived politicization.

The non-epistemic values guiding the protocol, such as environmental protection, global public health, and sustainable development, were shaped through extensive international deliberation. This process involved diverse stakeholders, including governments, scientists, industry leaders, and non-governmental organizations. These values reflected a broad societal

consensus on the importance of addressing ozone depletion, derived from transparent and participatory processes.

Once these values were democratically legitimized, Douglas's inquiry norms ensured their implementation adhered to epistemic rigor within the space of scientific inquiry. The research supporting the protocol followed the norm of clear and public restraints by explicitly defining the scope of inquiry and establishing clear guidelines for acceptable chemical use. Furthermore, the protocol adhered to the norm of criticism and response by fostering open-ended debate and revising its regulations as new scientific evidence emerged. For instance, subsequent amendments to the treaty incorporated additional substances as scientific understanding of their impact on the ozone layer evolved.

The norm of fostering diversity was another key strength of the Montreal Protocol's scientific foundation. Researchers from multiple disciplines and nations collaborated to model ozone depletion and assess the efficacy of proposed policy interventions. The inclusion of both developed and developing nations ensured that the treaty accounted for global equity, offering financial and technological assistance to help low-income countries transition away from ozone-depleting substances (Desombre 2000). This collaborative and inclusive approach contributed to the treaty's broad acceptance, making it both scientifically credible and socially equitable. As a result, the Montreal Protocol achieved near-universal adoption and delivered tangible environmental benefits, becoming a model for how science and international cooperation can address global challenges effectively.

Case Study 4: Does Not Look Politicized, But Is Politicized

The development of dietary guidelines in the mid-20th century illustrates a case that does not appear politicized at face value. Research shaping dietary recommendations seems neutral and driven by scientific inquiry into public health. However, deeper analysis reveals that the sugar industry covertly influenced scientific outcomes, shaping dietary guidelines to prioritize its financial interests (Kearns 2016). This case highlights how the failure to separate political norms from scientific inquiry leads to illegitimate politicization, and how Lusk's deliberative democracy and Douglas's inquiry norms could have averted such failures.

Lusk emphasizes the need for democratic selection of non-epistemic values to guide scientific research. In this case, public health should have been the central value shaping dietary research and guidelines. However, the deliberative process was undermined by the sugar industry's covert funding of studies that deflected blame for health issues like cardiovascular disease away from sugar and onto dietary fat (Kearns 2016).

Douglas's norms of inquiry, particularly the requirement for clear and public restraints, were also violated in this case. The sugar industry's influence was hidden, operating through undisclosed funding and biased research design. The lack of transparency in these constraints allowed the industry to exert undue influence over the direction and interpretation of dietary science. This failure was compounded by a violation of the norm of criticism and response, as alternative perspectives challenging the role of sugar in health risks were marginalized or ignored.

By failing to meet these standards, the sugar industry's influence on dietary guidelines represents a clear case of hidden politicization. The research appeared neutral and driven by public health priorities, but in reality, it reflected the interests of a powerful economic actor. This case makes clear the importance of maintaining both democratic legitimacy in value selection and

epistemic integrity in scientific inquiry to ensure that science serves society rather than special interests.

VI. Concluding Remarks

Looking ahead, the dual approach of Lusk's deliberative democracy and Douglas's inquiry norms holds significant potential for addressing contentious scientific domains. These fields involve complex epistemic questions and high societal stakes, often making them targets for politicization. Future research should focus on adapting this framework to diverse contexts, investigating how it can guide value selection, safeguard epistemic integrity, and rebuild public trust in science.

By bridging democratic legitimacy with epistemic rigor, this approach offers a practical roadmap for reconciling the demands of science and society. In an era where science is both indispensable and contested in shaping policy, the reconciliation of epistemic and democratic norms is not merely a philosophical aspiration but an urgent necessity for fostering informed and equitable decision-making.

Works Cited

- DeSombre, Elizabeth R. "The experience of the Montreal Protocol: Particularly remarkable, and remarkably particular." *UCLA Journal of Environmental Law and Policy*, vol. 19, no. 1, 2000, pp. 49–81, <https://doi.org/10.5070/l5191019217>.
- Douglas, Heather. "Differentiating scientific inquiry and politics." *Philosophy*, vol. 98, no. 2, Apr. 2023, pp. 123–146, <https://doi.org/10.1017/s0031819122000432>.
- Gannett, Lisa, "The Human Genome Project", *The Stanford Encyclopedia of Philosophy* (Fall 2023 Edition), Edward N. Zalta & Uri Nodelman (eds.), URL = <https://plato.stanford.edu/archives/fall2023/entries/human-genome/>
- Grey, Felix. "Corporate lobbying for environmental protection." *Journal of Environmental Economics and Management*, vol. 90, July 2018, pp. 23–40, <https://doi.org/10.1016/j.jeem.2018.03.008>.
- "Human Genome Project Fact Sheet." *Genome.Gov*, National Human Genome Research Institute, 13 June 2024, www.genome.gov/about-genomics/educational-resources/fact-sheets/human-genome-project.
- Kearns CE, Schmidt LA, Glantz SA. Sugar Industry and Coronary Heart Disease Research: A Historical Analysis of Internal Industry Documents. *JAMA Intern Med*. 2016 Nov 1;176(11):1680-1685. doi: 10.1001/jamainternmed.2016.5394. Erratum in: *JAMA Intern Med*. 2016 Nov 1;176(11):1729. doi: 10.1001/jamainternmed.2016.6774. PMID: 27617709; PMCID: PMC5099084.

Keren, Arnon. “Kitcher on well-ordered science: Should science be measured against the outcomes of ideal democratic deliberation?” *Theoria*, vol. 28, no. 2, 23 May 2013, pp. 233–244, <https://doi.org/10.1387/theoria.6590>.

Kitcher, Philip. 2001. *Science, truth and democracy*. Oxford: Oxford University Press.

Lin JC, Chang C, McCarthy MS, Baker-Butler A, Tong G, Ranney ML. Trends in Firearm Injury Prevention Research Funding, Clinical Trials, and Publications in the US, 1985-2022. *JAMA Surg*. 2024;159(4):461–463. doi:10.1001/jamasurg.2023.7461

Lusk, Greg. “Does democracy require value-neutral science? analyzing the legitimacy of scientific information in the political sphere.” *Studies in History and Philosophy of Science Part A*, vol. 90, Dec. 2021, pp. 102–110, <https://doi.org/10.1016/j.shpsa.2021.08.009>.

Omnibus Consolidated Appropriations Act, Pub. L. No. 104-208 (1996).

Rostron A. The Dickey Amendment on Federal Funding for Research on Gun Violence: A Legal Dissection. *Am J Public Health*. 2018 Jul;108(7):865-867. doi: 10.2105/AJPH.2018.304450. PMID: 29874513; PMCID: PMC5993413.